

Species-specific quantitative PCR (qPCR) protocol developed to identify *Haplosporidium pinnae*

1. Reagents:

- Specific primer pairs

Primer	Sequence (5'-3')	Base pars	Temperature annealing	Reference
HpF3	GCGGGCTTAGTTCAGGGG	165	60 °C	López-Sanmartín et al (2019)
HpR3	ACTTGTCCTTCCTCTAATAATAAGG			

- Thermo cycler Mx3000P Agilent
- PowerUp™ SYBR™ Green Master Mix (Applied Biosystems)

2. Mix:

- PowerUp™ SYBR™ Green Master Mix 5 µl
- HpF3 (10µM) 0.2 µl
- HpR3 (10µM) 0.2 µl
- DNA (100ng) 2 µl
- Water distilled and desionized until 10 µl

3. Amplification conditions:

- follow the manufacturer's instructions(https://assets.thermofisher.com/TFS-Assets/LSG/manuals/100031508_PowerUp_SYBRgreen_QRC.pdf)

Segments	Temperature	Time/segment	Num. of cycles
Segment 1	50°C	2 min	1
Segment 2	95 °C	2 min	1
Segment 3	95 °C	15 s	40
	60 °C	18 s	
	72 °C	1 min	
Segment 4	72 °C	15 s	1
	60 °C	18 s	
	95 °C	1 min	

3. Efficiency (E) was from the slope of the standard curve following formula:

- $E = 10^{-1/\text{slope} - 1}$

3. Melting curve was generated with temperatures increments of 0.5°C s⁻¹ starting at 60°C and ending at 95°C in order to ensure that a single PCR product was amplified for the primers.